Shared Control in pHRI: Integrating Local Trajectory Replanning and Cooperative Game Theory

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***Abstract*—In this article, we propose a two-stage shared control framework for physical human–robot interaction (pHRI) that addresses the inconsistency of human–robot commands and consider the influence of environmental information. In the human–robot–environment system, based on the human intention measured by the interaction force, autonomy will actively initiate the replanning when the human control intention is strong, generating a feasible local desired trajectory of the robot. At the same time, we define an index called predicted safety index (PSI) to measure the safety of the system status. When the human has control intention but does not reach the threshold, we propose a shared controller based on cooperative-game theory and PSI. Specially, it is designed within the model predictive control framework, utilizing cooperative game theory to analyze human–robot interaction behavior and treating the Pareto optimal solution as the control input. We conduct comparative experiments to evaluate the assistive performance of the proposed shared control algorithm through a waypoint tracking task with naive human users. User study with objective and subjective measures demonstrate that the algorithm effectively reduces human effort while maintaining tracking accuracy, thus enhancing both performance and safety.**

***Index Terms*—Physical human–robot interaction (pHRI), robot control, shared control (SHR).**

I. Introduction

A. Motivation

ALTHOUGH robots have made significant progress in perception, learning, and control, they cannot completely replace humans. Considering the complex and diverse task scenarios and various unexpected events encountered in practical applications, robot systems do not possess the capability to handle complex, unforeseen tasks in real time when operating completely autonomously. Integrating humans into the control loop, thereby enabling human–robot shared control (SHR), offers a novel approach to addressing these challenges. It can make full use of the complementary capabilities of both humans and robots in perception, decision-making, motion, and manipulation, enhancing the effect of cooperative task execution and promoting its application in many fields, such as telemedicine, rehabilitation, rescue, and exploration.

Physical human–robot interaction (pHRI) is a common form of human–robot interaction (HRI). Communication includes the following three types [1]: direct contact between the human and the robot, contact through a third object (e.g., comanipulation [2], cotransportation [3]), and HRI via a teleoperation device with force feedback (bilateral teleoperation [4], [5]). For many tasks in pHRI, especially comanipulation, conflicts of the human and the automated system are inevitable because the human and the automated system may complete the task in different ways. We consider the following scenario: when performing the subtask of obstacle avoidance, the automated system prepares to deflect left to avoid the obstacle based on the trajectory planning while the human may in the opposite direction for the same purpose due to more practical considerations. How should the robot behave in this situation?

In the field of SHR in pHRI, role arbitration between humans and robots is critical. For the abovementioned scenario, it is a common approach to treat humans and robots as a primary–secondary system. In a primary–secondary setup, the robot simply follows human movements, focusing on trajectory tracking [6], [7] or maintaining desired interaction forces [8]. This approach cannot strictly guarantee the security of the system when human operation level is limited, and increases the operational and cognitive burden of humans. SHR frameworks [9], [10], [11] that dynamically combine human and robot control weights have been proposed, but they often fail to align with human intentions in scenarios with significant discrepancies due to insufficient use of the robot's perception and planning capabilities. To address large disagreements, Medina et al. [12] treated them as prediction errors and calculates system input by solving a risk-sensitive optimal control problem that incorporates prediction uncertainty into the cost function. However, this method requires task-specific human data for predictive model training, increasing costs and reducing generality.

To resolve the human–robot conflict, we propose a two-stage SHR framework to handle the inconsistency of human–robot commands better. In the first stage, it is suitable for when the human–robot conflict is significant, the robot will replan and adjust its own reference trajectory according to the human's

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intention. For the scenario just mentioned, the robot will replan a trajectory with a local deflection. However, after adjustment in the first step, inconsistency of human–robot commands may still exist, which can be manifested as subtle differences. In other words, the humans and the robot are in a state of cooperation, and there is no “big difference” in intention between them. Still, the human may make subtle adjustments in execution that lead to a “small difference” in the desired trajectories of the human and the robot. In order to deal with the inconsistency at this stage, we propose a human–robot SHR strategy based on cooperative game theory, which uses the Pareto optimal solution as the control input to resolve human–robot conflict further and achieve safe and efficient interaction.

B. Related Work

To provide effective assistance during cooperation and reduce task burden, numerous SHR strategies have been proposed. According to whether the robot and the human are responsible for the same controlled variables, Li et al. [13] divided the existing SHR strategies into two categories: semiautonomous control and collaborative SHR. Since we primarily considers task commands from both humans and autonomy being directly input as control commands to the lower level motion controller to control the trajectory of the robot, we will mainly introduce the cooperative SHR. When the robot and the human achieve an equal relationship in controllable variables, they can each have their own cost function based on the task requirements. Depending on how the robot selects its own cost function, Losey et al. [14] classified the types of role allocation including primary–secondary, teacher–student, and collaboration.

The primary–secondary is the most classic and ubiquitous type of role allocation. The cost function of the robot is exactly the same as that of humans. The goal of the robot is to reduce the error of human expected task indicators while reducing the human task burden as much as possible. If the human in the human–robot system is regarded as the environment, a series of methods based on impedance control [8] or admittance control [15] can be applied, which makes the actual trajectory of the robot change according to human. However, as we discussed earlier, the limitations of a static primary–secondary paradigm are significant. Contrary to primary–secondary mode, in teacher–student arbitration, the robot act as the “teacher” to teach the human. The robot guides human involvement and gradually reduces its effort as the humans perform more independently. When it is applied in rehabilitation, it is usually manifested as an adjustment of impedance through setting threshold [16], designing the adaptive law [17], or learning-based methods [18]. For the collaboration role allocation, the human and the robot interact to achieve a common objective, and the relationship between them may dynamically change during task execution. The basis for the change depends on many factors, such as human trust, estimate the human state, and understanding of human intention. Many studies have attempted to develop a unified framework for human–robot cooperation [19], [20], [21], [22], since the equal partnership is most similar to humans performing tasks together.

It can be found that most of the works discussed above is to adjust the actual trajectory of the robot, and the desired trajectory of the robot has not changed, which increases the burden on humans making more efforts to achieve their goals. To address this problem, the method of deforming the future desired trajectory based on human’s interaction force is proposed in a pioneering work [23]. Zhou et al. [24] improved the trajectory deformation algorithm and apply it to the lower limb rehabilitation robot, ensuring that the desired trajectory after deformation remain within a reasonable range. However, these methods based on constrained optimization to deform the desired trajectory only considers the shape of the trajectory, ignoring its impact on the completion of the overall task and the surrounding environment.

Moreover, game theory is proved useful to analyze complex interactive behaviors involving two [25] or multiple agents [26], [27] in recent years. HRI can be regarded as a special two-agent game. Different from the multirobot game, for the human–robot system, only the behavior of the robot is controllable, while the human behavior is uncontrollable. Some research on game-theoretic SHR is applied in automated driving systems [28] to resolve driver–vehicle conflicts effectively. In [29], several paradigms (including noncooperative Nash, noncooperative Stackelberg, and cooperative Pareto) are discussed to model the steering interaction between the driver and vehicle in path-following control, and the corresponding equilibria are derived analytically. Ji et al. [30] further extended the game-based SHR framework by modeling the interaction in the presence of the uncertain driver behavior and the external disturbance. For the case of nonlinear interaction model, Li et al. [31] proposed a shared controller based on nonzero sum differential game and fast iterative solution method. For the research on interaction with the manipulator, the authors in [10] and [32] developed continuous role adaptation methods for human–robot SHR. An adaptation law is designed to adjust the robot cost function according to the human’s intention measured by the interaction force. Furthermore, the intentions of two agents can be estimated with the help of the observer [33]. Franceschi et al. [34] proposed a role arbitration framework that allows switching between cooperative and noncooperative game-theoretic models, applied to real-world collaborative carrying scenarios. However, the game-theoretic approaches only focus on the costs of both human and machine (determined by their respective efforts and goals), typically ignoring other environmental information (excluding the task), e.g., boundary conditions and obstacle distributions that determine system safety. In practical applications, the environment has an important influence on the system. In other words, a human–robot–environment system is supposed to be constructed, rather than a pure human–machine system. Thus, how to better process information from human, robot, and the environment in a unified framework is still an open problem.

C. Contributions and Article Structure

In this article, a two-stage SHR framework to better handle the inconsistency of human–robot commands and consider the influence of environmental information in a

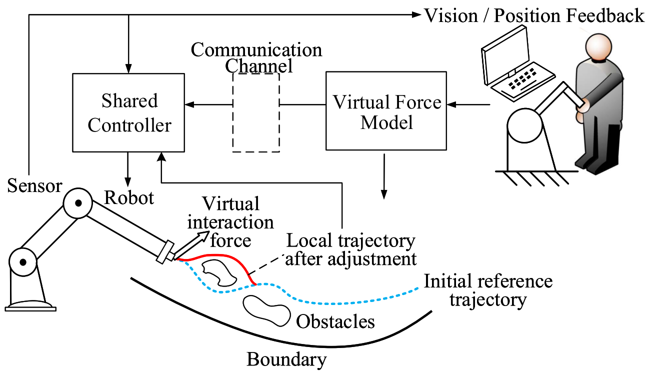

Fig. 1. System configuration.

human–robot–environment system. The main contributions of this article are as follows.

1) We propose an integrated framework where the robot adjusts both its desired and actual trajectories based on human intentions. This approach leverages the human's quick reactions and decision-making capabilities alongside the robot's high repeatability and continuous operation, reducing the cognitive load on the operator and avoiding long periods of tense teleoperation.
2) We introduce a local trajectory replanning strategy that uses a fast randomized motion planner to compute multiple feasible motion plans and selects the best plan based on the energy function that includes human adjustment efforts. This method ensures the feasibility of the modified trajectory and resolves initial conflicts between human and robot commands.
3) We design a shared controller based on cooperative-game theory to allow the human to adjust the actual trajectory. A safety index is designed and integrated into the optimization problem of solving Pareto optimal solutions to ensure the safety during the dynamic adjustment.

The rest of this article is organized as follows. Section II provides the problem statement. In Section III, we propose the method of reference trajectory replanning. Section IV presents the SHR strategy based on cooperative-game theory. The experimental results and user study are illustrated in Section V. Finally, Section VI concludes this article.

II. Problem Formulation and Algorithm Overview

A. Problem Formulation

The configuration of the system is shown in Fig. 1. In the tripartite system of the human–robot–environment, the robot's movement is influenced by both the autonomy-driven planning system and the human operator's inputs. The robot must adapt to changes in the environment and human commands, which are often inconsistent.

1) Initial Trajectory Planning: For a given desired position in the Cartesian space, the robot can automatically generate an optimal collision-free trajectory based on a series of planning algorithms considering the environmental information.

We take the method proposed in [35] as an example for a brief description. In essence, a set of feasible plans are computed, and the best plan is selected based on clinical criteria (such as shortest path, maximum clearance, etc.). Specifically, the initial and the target configuration of the robot in the Cartesian space are denoted by x_0 and x_{final}, respectively. The environmental information includes the positions of obstacles $o_i \in \mathcal{O}$ and boundaries $b_i \in \mathcal{B}$. To obtain the initial reference trajectory $x_{\text{dr}}(t)$ from x_0 to x_{final} with duration T (as shown by the blue dotted line), we first use one of sampling-based motion planning algorithms, rapidly exploring random tree (RRT) planners, to compute a set of feasible motion plans $p \in \mathcal{F}_0$ in 3-D environments with obstacles. Then, we select the best trajectory from the feasible trajectory set $\mathcal{F}_0$ as the initial reference trajectory $x_{\text{dr}}(t)$

$$x_{dr} = arg \min_{p\in\mathcal{F}_0} \left(\frac{\alpha_l}{T}\int_0^T |\dot{p}(t)| dt - \frac{\alpha_o}{T}\int_0^T \min_{\forall o_i\in\mathcal{O}} \|p(t)-o_i\|\, dt \right.$$
$$\left. - \frac{\alpha_b}{T}\int_0^T \min_{\forall b_i\in\mathcal{B}} \|p(t)-b_i\|\, dt \right). \tag{1}$$

It can be noticed that the motivated criterion is composed of three parts. The first term is to minimize the total path length. A shorter path length largely represents more effective motion with practical significance in various application scenarios. For example, it means less damage to tissue in surgery. The second and third term is to maximize the minimum clearance from obstacles and the boundary. Trajectories with a greater minimum clearance from obstacles and the boundary are safer because they are less prone to collision when deviations occur. $\alpha_l, \alpha_o, \alpha_b \in [0, 1]$ are all scalar used to adjust the weight of each criterion.

2) Human Intention Input: For the pHRI, the human can transmit force to the robot through a teleoperation device with force feedback or direct contact interaction. The interaction force can be regard as the expression of human intention. In remote operation scenarios, the inner control loop typically employs the following PID controller to output haptic feedback to the operator:

$$K_p(x_m - x_c) + K_i \int_0^t (x_m(\sigma) - x_c) d\sigma + K_d \dot{x}_m = -F_h \tag{2}$$

where $x_m \in \mathbb{R}^m$ and $x_c \in \mathbb{R}^m$ represent the current position and the central position of the end-effector in the frame of teleoperation device, respectively. K_p, K_i, and $K_d \in \mathbb{R}^{m\times m}$ are the proportional, integral, and derivative gains (diagonal and positive definite), respectively. $-F_h \in \mathbb{R}^m$ corresponds to the actual feedback force applied on operator during manipulation and F_h represents the virtual interaction force exerted by operator to the robot.

Faced with the contradiction between the instructions from the human's input and the optimal trajectory planned by autonomy, how should the control command be obtained at the current moment?

To clearly define our problem, the following assumptions are made.

Assumption 1: The robot operates in a known environment, with full knowledge of the obstacle positions and boundaries.

Assumption 2: The initial reference trajectory $x_{\mathrm{dr}}(t)$ is optimal and safe, capable of avoiding collisions.

Assumption 3: The robot can obtain accurate interaction forces in real time through a force feedback device.

Assumption 4: Humans can obtain the robot's position information in real time.

Problem: In a pHRI system, given the initial reference trajectory $x_{\mathrm{dr}}(t)$, a human operator provides real-time force input F_h to modify the robot's motion. The task is to design an SHR strategy that, based on the human's input, adjusts both the robot's desired trajectory and its actual motion in a way that minimizes the conflict between human and robot commands, while ensuring system safety and maintaining task performance.

B. Algorithm Overview

The control framework of our algorithm is shown in Fig. 1. Specifically, we propose a unified framework to handle human–machine relationships for pHRI efficiently. If pHRI never occurs, the robot will move along the initial reference trajectory driven by autonomy. When the human physically interacts with the robot, we first decide whether to adjust the robot reference trajectory based on the intensity of human intent, which is quantified by virtual interaction force. When the human intention is strong, the robot will try to replan local reference trajectory based on interaction force to reduce human–robot conflict. After that, in the face of "small difference" between human–machine commands, the robot will no longer adjust its desired trajectory but only adjust the actual trajectory through the proposed shared controller. Specifically, combined with the security of the system, we design a shared controller based on cooperative game theory to resolve human–robot conflict further and achieve safe and efficient interaction. This framework integrates the adjustment of both the current state and future state of the robot under the action of the human. On one hand, the adjustment of the future trajectory can reduce human effort and give more play to the advantages of autonomy in planning and precise control. On the other hand, instead of replanning the trajectory of the robot at every moment, only the actual behavior is controlled when the human intention is not strong, reducing the computational burden and improving the system efficiency.

III. Reference Trajectory Replanning

In this section, we will introduce how to adjust the local trajectory to better cater to human intention and reduce human effort.

When the human physically interacts with the robot, we first need to judge whether the human has a strong intention based on the virtual interaction force. Specifically, we consider the magnitude of the interaction force. When it exceeds a certain threshold, it indicates that the expectation conflict between the human and the robot is relatively large. Denote δ_f as the threshold of virtual interaction force F_h, if $||F_h|| \leq \delta_f$, it means that the desired command of the human has a little deviation from that of the robot, so the robot does not adjust its trajectory. On the contrary, if $||F_h|| > \delta_f$, the robot will reshape or replan the desired trajectory.

Although the initial reference trajectory $x_{\mathrm{dr}}(t)$ is to be replanned, the total time duration of the trajectory will not be changed. The updated local trajectory is guaranteed to be continuous with the original one. To start with, we define t_s and t_f as the initial time and end time of local trajectory, respectively. t_s is precisely the time when replanning starts and the t_f is determined by the predefined length of locally replanned trajectory. To accurately judge the operator's intent and avoid false triggers caused by noise and inadvertent actions, trajectory replanning was not immediately triggered when the interaction force exceeded the threshold. Considering the significant noise in force measurements, we applied a low-pass filter with a 10 Hz cut-off frequency to the interaction force measurements to reduce noise. In addition, we limited the maximum frequency of replanning to prevent excessive replanning f_p. Replanning was triggered only when the filtered interaction force exceeded the threshold and persisted for a specified duration $1/f_p$. The time range of the trajectory to be adjusted is $t \in [t_s, t_f]$, and the local trajectory is then updated as $x_{\mathrm{dr}}(t) = \tilde{x}_{\mathrm{dr}}(t)$. When $t \notin [t_s, t_f]$, since the condition for trajectory replanning has not been triggered, and the desired trajectory of the robot is still the original one $x_{\mathrm{dr}}(t)$. For local trajectory to be replanned, we first discrete $x_{\mathrm{dr}}(t)$ into N waypoints along $x_{\mathrm{dr}}(t)$, which is performed separately for each dimension. Denote γ_{di}^* as the discrete result of the i_{th} dimension

$$\gamma_{di}^* = [x_{di}(t_s), x_{di}(t_s+\delta), x_{di}(t_s+2\delta), \ldots, x_{di}(t_f)] \in \mathbb{R}^N \tag{3}$$

where $x_{di}, i = 1, 2, \ldots, m$ represent the i_{th} components of x_{dr}. δ is the selected discrete-time interval between consecutive waypoints. The number of waypoints is defined as $N = [(t_f - t_s)/\delta] + 1$.

Then, we discuss how to determine the updated waypoints of local trajectory. In order to ensure the positional continuity of the trajectory, it is known that $\tilde{x}_{\mathrm{dr}}(t) = x_{\mathrm{dr}}(t)$ at the start time t_s and the end time t_f. Similarly, RRT planner is also employed to generate a set of feasible motion plans from $x_{\mathrm{dr}}(t_s)$ to $x_{\mathrm{dr}}(t_f)$. Define $\gamma_d^* = [\gamma_{d1}^*, \ldots, \gamma_{dm}^*] \in \mathbb{R}^{N\times m}$, each feasible local motion plan consists of N updated waypoints $\gamma_d = [\gamma_{d1}, \ldots, \gamma_{dm}] \in \mathbb{R}^{N\times m}$. In a specified time interval, the set $\mathcal{G}$ formed by all γ_d is called locally feasible trajectories set. To choose the optimal trajectory from the locally feasible trajectories set, the energy function that maps trajectory adjustment to their relative costs need to be developed. Similar to [23], for $\forall \gamma_d \in \mathcal{G}$, we define the energy function after adjustment using waypoints as follows:

$$\begin{aligned} E(\gamma_d) &= E(\gamma_d^*) + \sum_{i=1}^{m} (\gamma_{di} - \gamma_{di}^*)^{\mathrm{T}} \left(-\hat{F}_{hi}\right) \\ &\quad + \sum_{i=1}^{m} \frac{1}{2\alpha} (\gamma_{di} - \gamma_{di}^*)^{\mathrm{T}} R (\gamma_{di} - \gamma_{di}^*) \end{aligned} \tag{4}$$

where $E(\gamma_d^*)$ is the energy of the initial local trajectory γ_d^*, and α is a positive scalar. Although the interval of the trajectory to be replanned is from t_s to t_f, whether to replan is judged based on the interaction force obtained at t_s. Therefore, it is impossible to obtain the interaction force from $t_s + \delta$ to t_f when the local trajectory is adjusted. The primary role of F_h is to calculate the energy after adjustment and to select the optimal local trajectory based on this energy. After applying filters and determining the interaction force over a short duration, $F_h(t_s)$ adequately reflects the intent of the human, making it reasonable to use for trajectory adjustment calculations. In other words, if there were no replanning algorithm, the force would likely remain close to $F_h(t_s)$ within a short period. To this extent, we adopt a simple prediction method that future applied forces will be the same as the current input, which involves taking the interaction force at the initial moment $F_h(t_s)$ as the prediction of $F_h(t), t \in (t_s, t_f]$, i.e., $\hat{F}_{hi} = \vec{1}F_{hi}(t_s) \in \mathbb{R}^N$, where F_{hi} is the i_{th} component of F_h, and $\vec{1} \in \mathbb{R}^N$ is a column vector with all ones. From (4), it can be observed that the energy function of adjusted trajectory consists of three parts: the first is the energy function of original local trajectory; the second is the work done by the trajectory adjustment to the human; and the third is the squared norm of $\gamma_{di} - \gamma_{di}^*$ with respect to matrix R. It can be used to ensure the trajectory after adjustment is natural to the human if a reasonable matrix $R \in \mathbb{R}^{N\times N}$ is chosen. According to [23], the minimum-jerk model is employed to describe the trajectory of human's movements. Thus, R is defined as follows:

$$R = A^T A, A = \begin{pmatrix} 1 & 0 & 0 & & 0 \\ -3 & 1 & 0 & & 0 \\ 3 & -3 & 1 & \cdots & 0 \\ -1 & 3 & -3 & & 0 \\ 0 & -1 & 3 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ 0 & 0 & 0 & & -3 \\ 0 & 0 & 0 & \cdots & 3 \\ 0 & 0 & 0 & & -1 \end{pmatrix} \in \mathbb{R}^{(N+3)\times N}. \tag{5}$$

To determine the "optimal" trajectory γ_d from the feasible trajectory set $\mathcal{G}$, we select by minimizing the energy of the adjusted trajectory. Since the adjusted trajectory energy contains an invariant γ_d^*, the optimization problem can be described independently by γ_d as follows:

$$\begin{aligned}\gamma_d &= \underset{\gamma_d \in \mathcal{G}}{\arg\min}\, E(\gamma_d) \\ &= \underset{\gamma_d \in \mathcal{G}}{\arg\min} \sum_{i=1}^{m} \left(\frac{1}{2\alpha} {\gamma_{di}}^T R \gamma_{di} - \left(\frac{1}{\alpha} {\gamma_{di}^*}^T R + {\hat{F}_{hi}}^T \right) \gamma_{di} \right).\end{aligned} \tag{6}$$

In this way, the reference trajectory of the robot γ_d after replanning is obtained, which not only considers human's intention but also ensures the operational safety. The method of actively adjusting expectations of autonomy to cater to the human can reduce human effort and lead to better user experience.

Remark 1: It is important to note that the assumption of constant interaction force within $[t_s, t_f]$ is only valid during the calculation of the energy after adjustment. After a trajectory adjustment, the system continues to sample and evaluate the interaction force for subsequent replanning. If the operator is dissatisfied with the adjusted trajectory, they can exert interaction force again, potentially triggering the next replanning. This means that interaction force sampling and evaluation are conducted in real time within $[t_s, t_f]$, and the assumption is only applicable during the energy function calculation.

IV. SHR Strategy Based on Cooperative-Game Theory

Although the contradiction between the human and robot has been preliminarily resolved after the autonomy adjusts its desired trajectory, there are still situations where the human intervention fails to meet the conditions for replanning the desired trajectory of the robot. For example, the human may want to make minor corrections to the trajectory. In this case, the human and the robot can be seen as having a cooperative relationship. A control strategy based on cooperative-game theory is developed in this section, where the actual behavior of the robot instead of the future one is controlled according to human input and system security.

A. System Safety Index

When the human and the robot jointly perform a task, the human may pay more attention to a certain subgoal while ignoring other underlying vital factors, such as safety. In addition, operators have varying levels of proficiency, and inexperienced operators may struggle to consider multiple factors simultaneously. For instance, in robotic surgery, a novice surgeon may focus more on the precision of the cut without fully considering the optimal posture of the surgical instrument, which may damage other soft tissue or cause other instruments to collide with each other. This situation is common in different application scenarios, such as rehabilitation training and codriving. Consequently, it is essential to develop a control strategy that continuously monitors and ensures the system's safety, thereby preventing potential dangers arising from operator overload. This control strategy should not only assist the operator in achieving their immediate goals but also maintain an overarching focus on safety and efficiency.

We first design an index to measure the safety of the system. Although the requirements for different human–machine systems differ, a generic expression can be defined through working conditions. Specifically, at time $k-1$, the robot reference trajectory position is $\gamma_d(k-1)$. From the perspective of system safety, we need to know how far the robot can deviate from $\gamma_d(k-1)$ without colliding with obstacles or boundaries, whose length is defined by the largest residual d_{res} as follows:

$$\begin{aligned} &d_{\text{res}}(k) \\ &= \min\left(d_0, \min_{\forall o_i \in o} \|\gamma_d(k-1) - o_i\|, \min_{\forall b_i \in B} \|\gamma_d(k-1) - b_i\| \right) \end{aligned} \tag{7}$$

where d_0 is the maximum allowed deviation without considering environment, which can be regarded as the permissible operational range. The significance of setting d_0 lies in the fact that

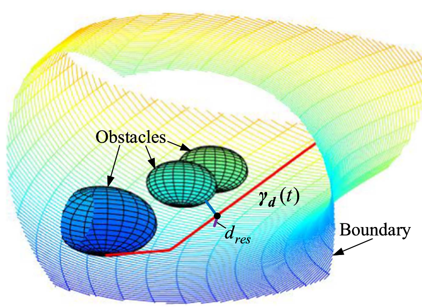

Fig. 2. Description of the largest residual. The red line is the desired trajectory of the robot generated by the autonomy. At time $k-1$, $\gamma_d(k-1)$ is denoted by the black point, whose minimum distance from obstacles and the boundary are represented by the blue line and the purple line, respectively. Then, the largest residual $d_{\text{res}}(k)$ is the length of the shorter of the two line segments and threshold d_0.

we do not want the actual deviation from the original trajectory to be excessive.

The description of the largest residual d_{res} is shown in Fig. 2. The actual deviation is represented by d, which is the Euclidean norm of the current position and the corresponding reference value

$$d(k) = \|x(k-1) - \gamma_d(k-1)\| . \tag{8}$$

Essentially, the condition to ensure the safety of the system is $d(k) \le d_{\text{res}}(k)$. But this is actually a "soft constraint." In other words, even if this working condition is violated, it does not mean that the system is unsafe and collisions must occur, but conversely, if this condition is met, the safety of the system can be ensured. We assume the initial reference trajectory is optimally planned to avoid obstacles and ensure safety. This assumption is supported by the availability of many advanced planning algorithms that can be used for different task scenarios. Thus, the closer the current position is to the reference trajectory (that is, the smaller the offset d), the safer the system is. Based on the abovementioned analysis, we define an indicator to quantify the security of the system, which is called the predicted safety index (PSI) and is denoted as follows:

$$\lambda(k) = \frac{\sqrt{d_{\text{res}}^2(k) - d_{\text{sat}}^2(k)}}{d_{\text{res}}(k)} \in (0,1). \tag{9}$$

Therein, $d_{\text{sat}} : [0, d_{\text{res}}] \to (0, d_{\text{res}})$ is a mapping constructed based on Richards curve

$$d_{\text{sat}}(k) = \frac{a_1 d_{\text{res}}(k)}{\{1 + \eta \cdot \exp[-\mu(d_{\max}(k) - a_2 d_{\text{res}}(k))]\}^{\frac{1}{\eta}}} \tag{10}$$

where $d_{\max}(k) = \min\{d(k), d_{\text{res}}(k)\}$. a_1, a_2, μ, η are parameters of the Richards curve, whose values determine the shape of the curve. They are predetermined values set according to the actual situation. Specially, a_1 is a positive constant set close to and less than 1, which determines the position of the upper asymptote. a_2 is a positive constant and determines the lag length before the curve rises. μ and η are all positive constants, and they respectively assess the rate of rise and degree of curvature of the curve. The value range of PSI is from 0 to 1, and the closer the value is to 1, the higher the safety of the current system state is. This metric can be used to adjust the weight of human participation subsequently.

B. Human–Robot Model

The dynamic model of the human–robot system is

$$\begin{aligned} &M(q)\ddot{q}(t) + \left[\frac{1}{2}\dot{M}(q) + C(q,\dot{q})\right]\dot{q}(t) + G(q) \\ &= \tau + {J_m}^T(q)F_h(t) \end{aligned} \tag{11}$$

where $M(q) \in \mathbb{R}^{n\times n}$ is the positive-definite and symmetric inertia matrix, $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ is a skew-symmetric matrix satisfying $\dot{q}^T C(q,\dot{q})\dot{q} = 0$ for all $q, \dot{q} \in \mathbb{R}^n$, $G(q) \in \mathbb{R}^n$ represents the gravitational force, and $\tau \in \mathbb{R}^n$ is the joint torque input. $J_m(q) \in \mathbb{R}^{m\times n}$ is the standard robot Jacobian matrix of the kinematic chain.

According to the method of two-loop impedance control in [10], the inner loop can be assumed to be perfect since the joint position control of the commercial manipulator is of high precision. If $q_r(t)$ is the reference position command, then $q(t) = q_r(t)$. The outer loop based on the impedance model in the Cartesian space is considered as

$$M_x\ddot{x}_r(t) + C_x\dot{x}_r(t) = u_r(t) + F_h(t) \tag{12}$$

where $M_x \in \mathbb{R}^{m\times n}$ and $C_x \in \mathbb{R}^{m\times m}$ represent the inertia and damping matrix, respectively. u_r is the control input in the Cartesian space. x_r is the reference position of the end-effector in the Cartesian space. Since we have assumed that $q(t) = q_r(t)$, and $\dot{x}_r(t) = J_m(q)\dot{q}_r(t)$, $x(t) = x_r(t)$. Then, we have

$$M_x\ddot{x}(t) + C_x\dot{x}(t) = u_r(t) + F_h(t). \tag{13}$$

For the convenience of subsequent analysis, we rewrite (13) in the form of a state equation as follows:

$$\begin{aligned} \dot{w}(t) &= Aw(t) + B_r u_r(t) + B_h F_h(t) \\ z(t) &= Cw(t) \end{aligned} \tag{14}$$

where

$$\begin{aligned} &w = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}, A = \begin{bmatrix} 0_{m\times m} & I_{m\times m} \\ 0_{m\times m} & -M_x^{-1}C_x \end{bmatrix} \\ &B_r = B_h = \begin{bmatrix} 0_{m\times m} \\ M_x^{-1} \end{bmatrix}, C = \begin{bmatrix} I_{m\times m} & 0_{m\times m} \end{bmatrix}. \end{aligned} \tag{15}$$

C. Game Controller Design

After the robot has performed local trajectory replanning, the human may still participate in the interaction through a teleoperation device with force feedback. When the interaction cannot reach the threshold of replanning at this time, a shared controller based on cooperative game theory is applied. The SHR strategy that coordinates commands from both the human and the autonomy is developed based on the distributed model predictive control (MPC) approach. Thus, we first discretize the continuous-time system (14) at a sample time T, the discrete-time system can be obtained as follows:

$$\begin{aligned} w(k+1) &= A_d w(k) + B_{rd} u_r(k) + B_{hd} F_h(k) \\ z(k) &= Cw(k) \end{aligned} \tag{16}$$

where

$$A_d = e^{AT}, B_{rd} = B_r \int_0^T e^{As} ds, B_{hd} = B_h \int_0^T e^{As} ds. \tag{17}$$

Assume that both the human and the autonomy will use the next N_p state of the robot as their targets. In other words, the human expectations at time step k are expectations for a preview time in the future, and the preview time will vary for different people or in different situations. If the preview time is T_p, the number of preview states is $N_p = T_p/T$. Accordingly, the prediction time of the autonomy is consistent with the human. Denote $X_{dh}(k)$ and $X_{\mathrm{dr}}(k)$ as the vectors composed of the desired states of the human and the robot at $k+1$ step to $k+N_p$ step, respectively

$$X_{dh}(k)=\begin{bmatrix} x_{dh}(k+1) \\ x_{dh}(k+2) \\ \vdots \\ x_{dh}\left(k+N_p\right) \end{bmatrix}, X_{\mathrm{dr}}(k)=\begin{bmatrix} x_{\mathrm{dr}}(k+1) \\ x_{\mathrm{dr}}(k+2) \\ \vdots \\ x_{\mathrm{dr}}\left(k+N_p\right) \end{bmatrix}. \tag{18}$$

In the abovementioned equation (18), the desired states of the human were derived from the discrete state-space equation (16), assuming constant interaction force during the preview time

$$\begin{aligned} x_{dh}(k+i) &= \begin{cases} x_{\mathrm{dr}}(k+i) & \|F_h(k)\| = 0 \\ Cw_h(k+i) & \|F_h(k)\| \neq 0 \end{cases} \\ w_h(k+i) &= A_d w_h(k+i-1) + B_{hd} F_h(k) \\ w_h(k) &= w(k) \end{aligned} \tag{19}$$

where $i = 1, \ldots, N_p$.

When modeling the behavior of the human in the process of participating in the interaction, a method based on cooperative game theory is adopted. The relationship between the human and the autonomy can be regarded as a cooperative relationship, and the strategy of each agent takes into account the costs of both agents. Because the difference between the target trajectories of the human and the autonomy is small, and the human does not show a strong intention to take control but needs to make small adjustments to the actual trajectory of the robot, it is reasonable to model human behavior with cooperative games. For the cooperative game, "Pareto equilibrium" refer to a situation where neither agent can unilaterally change its strategy to gain more, which corresponds to the "Nash equilibrium" in the classic noncooperative game theory. To derive the analytical expressions of the human's and the autonomy's strategies in the Pareto equilibrium by the distributed MPC approach, we first construct the cost functions of both agents. The tracking errors is denoted by $E_h(k)$ and $E_r(k)$ as follows:

$$\begin{aligned} E_h(k) &= Z(k) - \ X_{dh}(k) \in {}^{mN_p} \\ E_r(k) &= Z(k) - X_{\mathrm{dr}}(k) \in {}^{mN_p} \end{aligned} \tag{20}$$

where

$$Z(k) = \begin{bmatrix} z(k+1) \\ z(k+2) \\ \vdots \\ z\left(k+N_p\right) \end{bmatrix}.$$

The derivation of $Z(k)$ follows the HRI dynamics equations (16) by iterating N_p steps. When constructing the cost function based on cooperative game theory, the human and the autonomy consider each other's interests, which makes the cost functions include the tracking error of both agents. Importantly, we also take the impact of surrounding environmental factors on system security into consideration. Specifically, PSI designed in the last section is used to adjust the error weight matrix of the human and autonomy to achieve dynamic authority allocation. The ratio between the human and the autonomy grows with the increase of the assessed system safety. When the robot runs under a higher PSI, the target of the human should be considered more; when the safety of the system is low, the autonomy should occupy more control authority in the SHR system. The cost functions of the human and the autonomy are presented, respectively

$$\begin{aligned} J_h(k) &= E(k)^\top Q(k) E(k) + (1-\lambda(k)) U_h^\top(k) U_h(k) \\ J_r(k) &= E(k)^\top Q(k) E(k) + \lambda(k) U_r^\top(k) U_r(k) \end{aligned} \tag{21}$$

where

$$\begin{aligned} E(k) &= \begin{bmatrix} E_h(k) \\ E_r(k) \end{bmatrix}, Q(k) = \begin{bmatrix} \lambda(k) Q_h & 0 \\ 0 & (1-\lambda(k)) Q_r \end{bmatrix} \\ Q_h &= \mathrm{diag}\{Q_{h1} \ \cdots \ Q_{hN_p}\} \\ Q_r &= \mathrm{diag}\{Q_{r1} \ \cdots \ Q_{rN_p}\} \\ Q_{hi} &= \mathrm{diag}\{q_{h1} \ \cdots \ q_{hm}\}, i = 1, \ldots, N_p \\ Q_{ri} &= \mathrm{diag}\{q_{r1} \ \cdots \ q_{rm}\}, i = 1, \ldots, N_p. \end{aligned} \tag{22}$$

U_h and U_r are control actions of the human and the autonomy from k step to $k+N_p-1$ step, respectively

$$U_h(k)=\begin{bmatrix} F_h(k) \\ F_h(k+1) \\ \vdots \\ F_h\left(k+N_p-1\right) \end{bmatrix}, U_r(k) = \begin{bmatrix} u_r(k) \\ u_r(k+1) \\ \vdots \\ u_r\left(k+N_p-1\right) \end{bmatrix}. \tag{23}$$

Then, we discuss how to establish the global prediction equation (GPE), including the preview. By iterating the HRI dynamics equations (16) N_p steps, $Z(k)$ can be derived. Define $Z_g(k)$ as the global state, and the GPE is presented as follows:

$$\begin{aligned} Z_g(k) &= \psi_g w(k) + \theta_{hg} U_h(k) + \theta_{rg} U_r(k) \\ \psi_g &= \begin{Bmatrix} \psi \\ \psi \end{Bmatrix}, \theta_{hg} = \begin{Bmatrix} \theta_h \\ \theta_h \end{Bmatrix}, \ \theta_{rg} = \begin{Bmatrix} \theta_r \\ \theta_r \end{Bmatrix} \\ \psi &= \begin{bmatrix} CA_d \\ CA_d{}^2 \\ \vdots \\ CA_d{}^{N_p} \end{bmatrix}, Z_g(k) = \begin{bmatrix} Z(k) \\ Z(k) \end{bmatrix} \in \mathbb{R}^{2mN_p} \\ \theta_h &= \begin{bmatrix} CB_{hd} & 0 & \cdots & 0 \\ CA_d B_{hd} & CB_{hd} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA_d{}^{N_p-1} B_{hd} & CA_d{}^{N_p-2} B_{hd} & \cdots & CB_{hd} \end{bmatrix} \end{aligned}$$

$$\theta_r = \begin{bmatrix} CB_{rd} & 0 & \cdots & 0 \\ CA_d B_{rd} & CB_{rd} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA_d{}^{N_p-1} B_{rd} & CA_d{}^{N_p-2} B_{rd} & \cdots & CB_{rd} \end{bmatrix}. \tag{24}$$

The next step is to derive the Pareto solution to get the SHR input. The predictive control problems to be solved for the human and the autonomy are as follows:

$$\min_{U_h} J_h(k) \text{ and } \min_{U_r} J_r(k)$$
$$\text{s.t.} Z_g(k) = \psi_g w(k) + \theta_{hg} U_h(k) + \theta_{rg} U_r(k). \tag{25}$$

$(U_h{}^*, U_r{}^*)$ is the Pareto equilibrium solution for the human–robot cooperative game-based SHR problem, and $U_r{}^*$ will be used for robot control input.

Denote the global target as $X_d = [X_{dh}^T \quad X_{dr}^T]^T$. We define the error terms associated with the human and the autonomy as follows:

$$\begin{aligned} \varepsilon_r(k) &= X_d(k) - \psi_g w(k) - \theta_{hg} U_h(k) \\ \varepsilon_h(k) &= X_d(k) - \psi_g w(k) - \theta_{rg} U_r(k). \end{aligned} \tag{26}$$

Then, we have

$$E = \theta_{rg} U_r(k) - \varepsilon_r(k) = \theta_{hg} U_h(k) - \varepsilon_h(k). \tag{27}$$

By combining (26) and (24), the cost functions (21) can be rewritten as

$$\begin{aligned} J_h(k) &= \left\| \begin{bmatrix} S_Q\left(\theta_{hg} U_h(k) - \varepsilon_h(k)\right) \\ \sqrt{1-\lambda}\, U_h(k) \end{bmatrix} \right\|^2 \\ J_r(k) &= \left\| \begin{bmatrix} S_Q\left(\theta_{rg} U_r(k) - \varepsilon_r(k)\right) \\ \sqrt{\lambda}\, U_r(k) \end{bmatrix} \right\|^2 \end{aligned} \tag{28}$$

where $S_Q^\top S_Q = Q(k)$. To solve the optimization problem (25) and obtain the Pareto solution, QR algorithm [36] can be employed. The optimal input of the human and the autonomy are calculated by

$$\begin{aligned} U_h^*(k) &= \begin{bmatrix} S_Q \theta_{hg} \\ \sqrt{1-\lambda} \end{bmatrix}^{-1} \begin{bmatrix} S_Q \\ 0 \end{bmatrix} \varepsilon_h(k) = L_h \varepsilon_h(k) \\ U_r^*(k) &= \begin{bmatrix} S_Q \theta_{rg} \\ \sqrt{\lambda} \end{bmatrix}^{-1} \begin{bmatrix} S_Q \\ 0 \end{bmatrix} \varepsilon_r(k) = L_r \varepsilon_r(k). \end{aligned} \tag{29}$$

It can be noticed that $U_h^*(k)$ and $U_r^*(k)$ cannot be solved merely by (29), since the control sequence of the other agent is needed. To solve this problem, the convex iteration approach [37] is employed. Construct the following auxiliary equations:

$$\begin{cases} U_h(k)^{[p+1]} = \rho U_h^*(k)^{[p]} + (1-\rho) U_h(k)^{[p]} \\ U_r(k)^{[p+1]} = (1-\rho) U_r^*(k)^{[p]} + \rho U_r(k)^{[p]} \end{cases} \tag{30}$$

where $\rho \in (0, 1)$ is the iteration weight, and p is the number of iterations. As $p \to \infty$, (30) becomes

$$\begin{cases} U_h(k)^{[\infty]} = \rho U_h^*(k)^{[\infty]} + (1-\rho) U_h(k)^{[\infty]} \\ U_r(k)^{[\infty]} = (1-\rho) U_r^*(k)^{[\infty]} + \rho U_r(k)^{[\infty]} \end{cases}. \tag{31}$$

By combing (31), (29), and (26), we can obtain

$$\begin{aligned} U_h(k)^{[\infty]} &= \underbrace{(I - L_h\theta_{rg}L_r\theta_{hg})^{-1}(H_h - L_h\theta_{rg}H_r)}_{K_{h*}} \begin{bmatrix} w(k) \\ X_d(k) \end{bmatrix} \\ U_r(k)^{[\infty]} &= \underbrace{(I - L_r\theta_{hg}L_h\theta_{rg})^{-1}(H_r - L_r\theta_{hg}H_h)}_{K_{r*}} \begin{bmatrix} w(k) \\ X_d(k) \end{bmatrix} \end{aligned} \tag{32}$$

where

$$H_h = \begin{bmatrix} -L_h\psi_g & L_h \end{bmatrix}, H_r = \begin{bmatrix} -L_r\psi_g & L_r \end{bmatrix}. \tag{33}$$

The sequence pair $(U_h(k)^{[\infty]}, U_r(k)^{[\infty]})$ is the Pareto equilibrium. From the definition of $U_r(k)$, we can observe that $U_r(k)^{[\infty]}$ has N_p elements from $u_r(k)^{[\infty]})$ to $u_r(k+N_p-1)^{[\infty]})$. However, the system only sends the control input to the robot at time step k. According to the concept of receding horizon, the first element of $U_r(k)^{[\infty]}$ is taken as the control input. Thus, the shared controller can be expressed as follows:

$$u_r(k) = U_r{}^*(k) = [1 \quad 0 \quad \cdots \quad 0] \; K_r{}^* \begin{bmatrix} w(k) \\ X_d(k) \end{bmatrix}. \tag{34}$$

Remark 2: When human has no interaction, the shared controller can be regarded as a controller pursuing the robot desired trajectory x_{dr}. Since x_{dr} is planned to avoid obstacles and ensure safety, the PSI is set to 1, i.e., $\lambda(k) = 1$. Under this condition, problem (25) be reformulated as a single objective optimization problem. By choosing the optimal value of this objective function as the Lyapunov candidate, it can be proved that the tracking error of robot desired trajectory is bounded. When human has interaction, the tracking error increases to accommodate the human's intention to adjust the trajectory locally. This results in a decrease in the PSI, which eventually approaches the boundary state, $\lambda(k) = 0$. Following similar steps as in the noninteraction case, it can be proven that the tracking error under the disturbance of interaction force is bounded. Therefore, the proposed SHR system is stable. Furthermore, the local trajectory replanning ensures the system safety. By selecting an appropriate triggering threshold δ_f through experiment, it is guaranteed that $d(k) \leq d_{\text{res}}(k)$ before replanning occurs.

V. Experiments

A. Experimental Setup

The experimental setup is illustrated in Fig. 3, where a manipulator with 7 degrees of freedoms (DOFs), KUKA LBR IIWA R800, was used as the experiment platform. The manipulator was under the mode of position control in the Cartesian space and an admittance controller was utilized to translate human interaction force and robot control input into position command. The transfer function of this admittance controller followed the Laplace form of the impedance model (13). In scenarios like teleoperation surgery, the robot must actively avoid obstacles and plan motion trajectories to navigate around biological tissues and organs, ensuring safety. To simulate such conditions, we placed some cubes as obstacles in the experiment. The real-time position of the cubes was tracked via the VICON motion capture

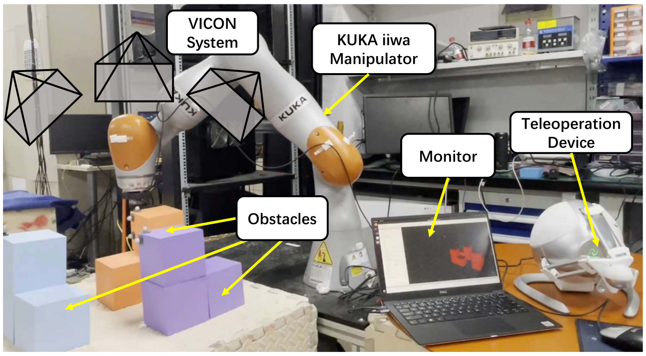

Fig. 3. Experimental setup. The operator controlled the end-effector of a 7-DOF KUKA iiwa manipulator using a 3-DOF haptic teleoperation device to navigate around obstacles. The VICON motion capture system tracked the positions of obstacles and the operator received the visual feedback via a laptop monitor.

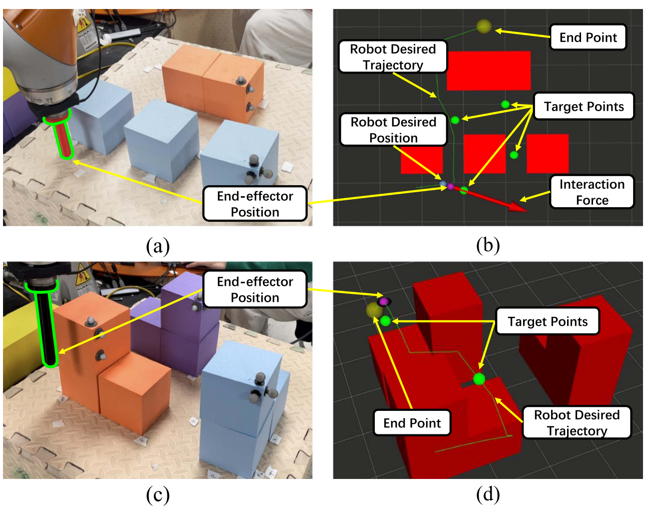

Fig. 4. Task scenarios. (a) and (c) Two-dimensional and three-dimensional configurations with a stick-likeend-effector. (b) and (d) Screenshots of the monitor with visual feedback.

system using reflective trackers attached to the cubes. In addition, a rectangular cuboid region was defined in the algorithm as the permissible operational area for the end-effector of robot. The operator used a 3-DOF haptic device, the Novint Falcon, to interact with the robot. A monitor provided the operator with visual feedback of the robot and the remote environment, offering real-time visualization of obstacle positions, the end-effector position, and the virtual interaction force. The control algorithm was running on robot operating system on a workstation (with Ubuntu 20.04 LTS, 7.7 GiB Memory and an Intel(R) Core(TM) i7-10710 U CPU @ 1.10 GHz).

Depicted in Fig. 4, the task scenario is the target pursuing and obstacle avoidance. Initially, the autonomy planed a collision-free trajectory with the shortest path from the start point to the end point. During the task, the operator was required to control the end-effector to pass target points as accurately as possible while avoiding obstacles, guided by visual feedback on the monitor. The operator can see the position of the end-effector

TABLE I
PARAMETERS USED IN EXPERIMENTS

Parameter	Value	Parameter	Value
K_p	$60I_{3\times3}$	a_2	0.2
K_i	$I_{3\times3}$	μ	200
K_d	$10I_{3\times3}$	η	0.1
δ_f	3.0 N	M_x	$I_{3\times3}$ kg
δ	0.1 s	C_x	$40I_{3\times3}$ kg/s
N	380	T	0.1 s
α	100	N_p	50
d_0	0.03 m	Q_{ri}	$1000I_{3\times3}$
a_1	1	Q_{hi}	$10000I_{3\times3}$

and obstacles. As they apply force via the stick, the magnitude and direction of the interaction force will also be displayed in the form of arrows. If the interaction force exceeds the predefined threshold δ_f, the force arrow on the monitor turns red, indicating that trajectory replanning has been triggered. In addition, to inform the operator of the decisions of robot, the desired trajectory and desired position of the autonomy will also be visualized on the monitor. Experiments were conducted in both two-dimensional and three-dimensional task spaces.

The parameters used in our algorithm are listed in Table I. Unless otherwise stated, the parameters in subsequent experiments will be as specified in Table I.

B. Task Performance

To validate the effectiveness of each stage of our algorithm, we conducted experiments under different dimensional configurations. The results demonstrate that our algorithm can reduce human effort while maintaining the accuracy of target pursuit. Details of the experiment analysis are as follows.

1) Two-Dimensional Task Space: The two-dimensional configuration was shown in Fig. 4(a) and (b), and the coordinates of the start and end points in the manipulator frame were $(0.38, -0.41)$ and $(0.55, 0)$, respectively. For the replanning part of our algorithm, we used the basic RRT algorithm to quickly generate a feasible trajectory. Four target points $P_i,\ i = 1, \ldots, 4$ were positioned around the initial trajectory at $(0.60, -0.20)$, $(0.48, -0.24)$, $(0.62, -0.33)$, and $(0.50, -0.42)$, all within the permissible operational area.

Before starting the task, the robot planned a feasible trajectory (as shown in the black line in Fig. 5) with the shortest path from start to end using a heuristic bidirectional RRT$*$ path planning algorithm and a Time-Elastic-Band trajectory planning algorithm. Fig. 5 shows the robot's planned trajectories during the task, including the direction of interaction forces at specific times. Fig. 6 (Top) displays the magnitude of interaction forces during the task. At 8.9 s, the interaction force exceeded the threshold, triggering local trajectory replanning. At the same time, the interaction force directed toward one of the target points (see Fig. 5, Left). The replanning was completed at 11.7 s, marked by a star in Fig. 6. It can be observed that the new desired trajectory of the robot closely aligns with human intention due to its proximity to the target points. Before reaching the first target point, the operator did not perform any actions, and the interaction force was zero.

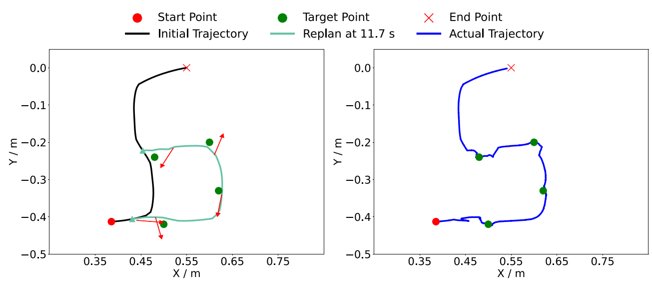

Fig. 5. Left: The desired trajectories of the robot. The start and end points of replanning are marked by triangles, and the directions of interaction force when the operator triggered replanning or started to deviate the end-effector from the planned trajectory are marked by red arrows. Right: The final actual trajectory of the robot with human interaction.

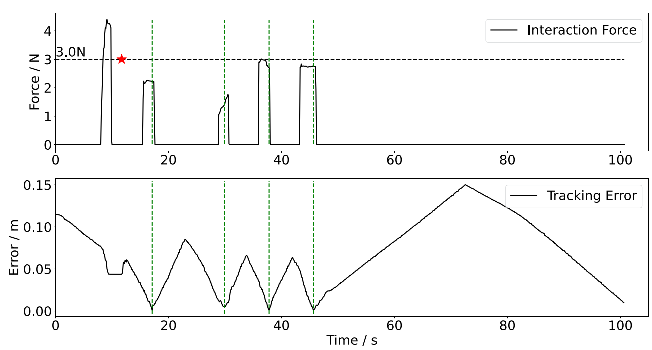

Fig. 6. Top: The magnitude of interaction force during the task. Trajectory replanning was triggered once, with a red star marking the completion time of replanning. Bottom: The position error, which is the minimum distance between the end-effector and the four target points plus the end point, during the task. The green dashed line marks the minimum tracking errors for each target point.

As the robot moved, the operator applied interaction forces when approaching each target point, causing the end-effector to deviate slightly from the autonomy's desired trajectory to better approach the target points. The position and the direction of interaction forces when the operator began interacting with the teleoperation device are also noted in Fig. 5 (Left). During this period, each interaction force remained below the replanning threshold, allowing the human to fine-tune the robot's actual trajectory. The final actual trajectory of the robot is shown in the blue line in Fig. 5, where it can be seen that it passed through the four target points and reached the endpoint. For a more accurate assessment of the results, Fig. 6 (bottom) shows the real-time minimum distance between the end-effector and the target points. Green dashed lines indicate the minimum tracking errors for the four target points, which were 0.0012 m, 0.0013 m, 0.0046 m, and 0.0010 m, respectively.

2) Three-Dimensional Task Space: The three-dimensional configuration was shown in Fig. 4(c) and (d), and the coordinates of the start and end points in the manipulator frame were $(0.38, -0.41, 0.17)$ and $(0.4, 0.05, 0.08)$, respectively. As in two-dimensional experiment, the robot planned a trajectory with the shortest path initially. Regarding the planning method adopted in the three-dimensional experiment, we used a heuristic bidirectional RRT$*$ algorithm for replanning and restricted the sample space of the RRT algorithm to a small cuboid region based on the interaction force direction at each replanning instance. These path planning optimizations reduce the variability of feasible trajectories, increasing the likelihood of generating a new trajectory that aligns with human intention. Two target points $P_i,\ i = 1, 2$ were positioned around the initial trajectory at $(0.50, -0.30, 0.18)$ and $(0.40, -0.07, 0.15)$, all within the permissible operational area.

Fig. 7 (Left and Middle) shows the initial planned and replanned trajectories during the task, including the direction of interaction forces at specific positions. Fig. 8 (Top) displays the magnitude of interaction forces during the task. At 14 s, the operator triggered local trajectory replanning. At the same time, the interaction force directed toward one of the target points (see Fig. 7, Left). The replanning was completed at 19.9 s. However, since only the local trajectory was replanned, the robot detoured despite the new desired trajectory being close to the target points. Consequently, at 52.9 s, the operator triggered local trajectory replanning again to shorten the remaining trajectory (see Fig. 7, Middle). Similar to two-dimensional experiment, the operator adjust the actual trajectory when the end-effector was near target points. The directions of interaction forces at the moments when the operator began interacting with the teleoperation device are also noted in Fig. 7. As the replanning was triggered twice, there are two star marked in Fig. 8 (Top).

Finally, the final actual trajectory passed through the target points and the end point as shown in Fig. 7 (Right). For the quantitative results, Fig. 8 (Bottom) presents the real-time minimum distance between the end-effector and the two target points plus the end point. Green dashed lines indicate the minimum tracking errors of the two target points, which are 0.0015 m and 0.0023 m, respectively.

C. Parameters Analysis

There are two key parameters in our algorithm that impact task performance: the length of the locally replanned trajectory, defined by the number of waypoints N, and the PSI, which quantifies the safety of system. To elaborate on the effects of these parameters, we conducted two comparative experiments as follows.

1) Length of Locally Replanned Trajectory: To illustrate the effects of replanning length N, we conducted a comparative experiment in a two-dimensional target pursuing task scenario as described above, where $N = 200, 380, 600$, respectively. Fig. 9 displays the robot's desired and actual trajectories in different replanning length N, while Fig. 10 shows the changes in the magnitude of interaction force and the minimum distance between the end-effector and the four target points plus the end point.

As N increases, the shape of the locally replanned trajectory becomes more uncertain. It can be found that usually only the initial part of the trajectory aligns with human intention, requiring the operator to replan again for the latter part. This results in more frequent replanning, leading to a longer completion time [see Fig. 10(c)] and more burden on the operator. As shown in Fig. 9(c), the second and third replanning instances were

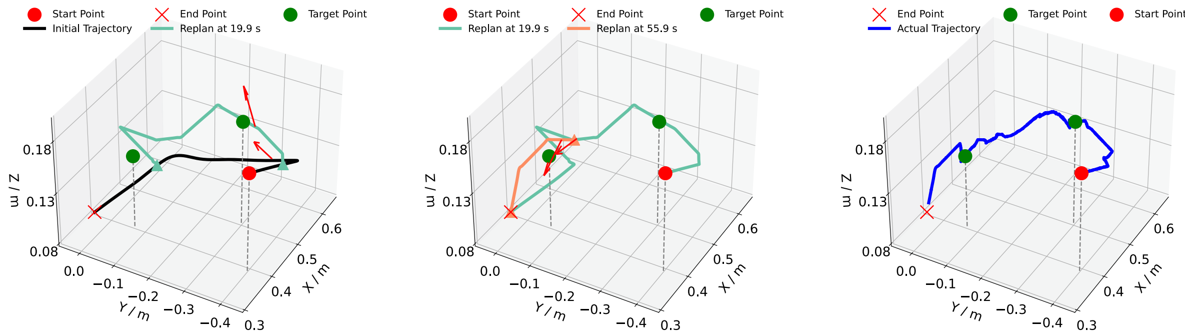

Fig. 7. Left and Middle: The desired trajectories of the robot. The start and end points of replanning are marked by triangles, and the directions of interaction force when the operator triggered replanning or started to deviate the end-effector from the planned trajectory are marked by red arrows. Right: The final actual trajectory of the robot with human interaction. Gray dash lines indicate the projections of both the target and start points onto the plane z=0.08 m.

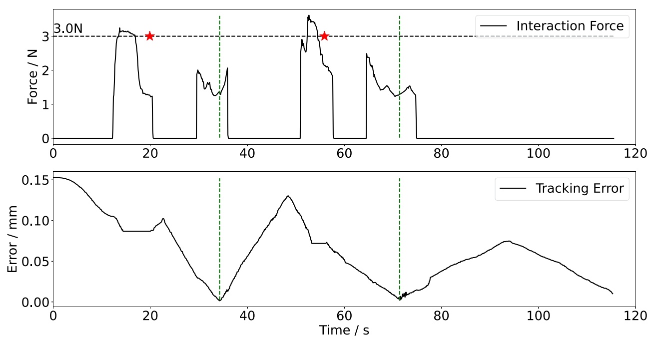

Fig. 8. Top: The magnitude of interaction force during the task. Trajectory replanning was triggered twice, with two red stars marking the completion times of replanning. Bottom: The position error, which is the minimum distance between the end-effector and the two target points plus the end point, during the task. The green dashed line marks the minimum tracking errors for each target point.

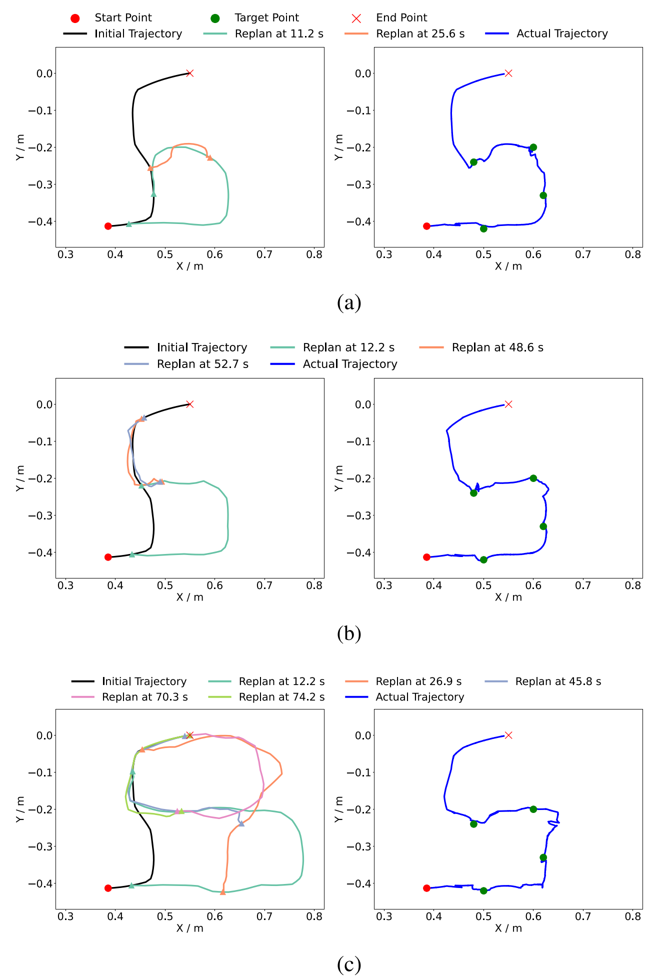

Fig. 9. Desired and actual trajectories of the robot with different lengths of locally replanned trajectories. Different replanned trajectories in one experiment are distinguished by different colors, and their start and end points are marked by triangles.

triggered because the latter part of the trajectory was far from the target points. In addition, since the result of the fourth replanning was entirely in the wrong direction, the operator triggered a fifth replanning immediately after the fourth. When N is too small, the velocity of robot increases steeply due to the constrained total duration of the replanned trajectory. It is clear in Fig. 10(a), where at around 42 s, the operator had already passed all target points. Although it shortens the completion time, the sudden change in velocity makes operation more difficult and reduces target-pursuing accuracy [see Fig. 9(a)]. Therefore, the length of the local replanned trajectory should be adjusted according to the scenario of the task, avoiding being too large or too small. In general, it can be selected to be comparable to the size of a single obstacle in the environment, which can increase the probability of the replanned trajectory avoiding an obstacle and be more in line with human intentions. In our scenario, it is $N = 380$ to minimize the number of replanning times and avoid abrupt changes in robot velocity.

2) Predicted Safety Index: To validate the effectiveness of the PSI, we conducted an experiment simulating scenarios where operators might issue risky control commands leading to collisions. The operator applied a force causing the robot to deviate from its initial trajectory and move closer to a circular obstacle. As trajectory replanning also contributes to ensuring system safety, we intentionally avoided triggering replanning during the experiment to independently verify the effect of PSI. Therefore, the magnitude of the interaction force remained below the replanning threshold, which is 3.7 N in this experiment. Other parameters maintained their values from Table I.

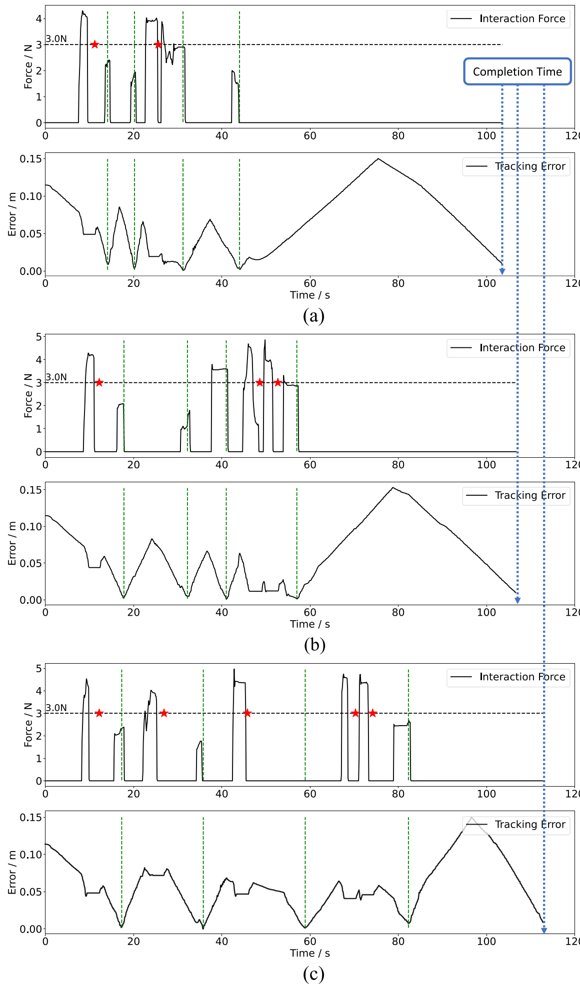

Fig. 10. Changes in magnitude of interaction force and real-time error (as described in Fig. 6) with different lengths of locally replanned trajectories. The red star marks the completion of replanning, and the green dashed line marks the minimum tracking errors for each target point. Completion times under different conditions are marked by blue dashed lines.

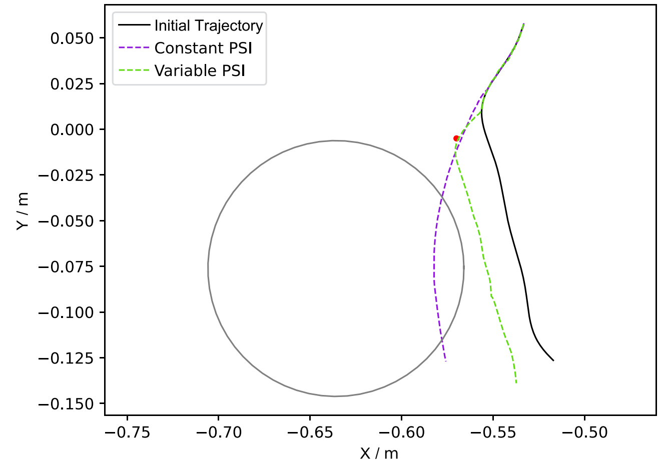

Fig. 11. Simulation experiment validating the efficiency of PSI. The black trajectory represents the initial planned trajectory. The green and purple trajectories correspond to the actual trajectories with variable PSI and constant PSI, respectively. The purple trajectory collides with the obstacle, while the green one does not.

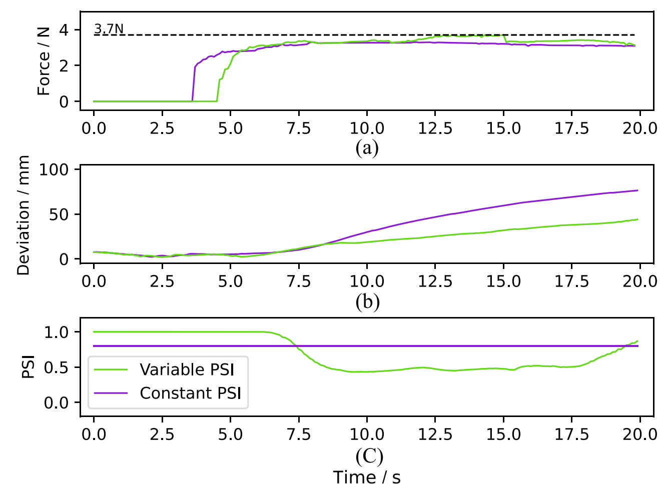

Fig. 12. (a) Interaction force, which is below the replanning threshold. (b) Deviation of the end-effector point from the desired point of the robot, along the initial planned trajectory. (c) Variation of PSI.

The actual trajectory of the end-effector with variable PSI is depicted as the green curve in Fig. 11. Despite continuous exertion of the interaction force (as the operator kept directing the robot toward the obstacle), the robot avoided collision. However, when PSI remained constant at a value of 0.8, as shown by the purple curve in Fig. 11, a collision occurred. Fig. 12 illustrates the changes in the magnitude of interaction force, deviation of the actual trajectory from the initial trajectory, and PSI values. As the end-effector approached an obstacle, the PSI value rapidly decreased, thereby reducing the weight of human intention in the cost function (21). These results indicate that the proposed PSI can autonomously assess the safety of the current environment, effectively preventing collisions between the robot and obstacles.

D. User Study

Human behavior exhibits inherent uncertainty, and although within the context of identical platform and task objective, different individuals may employ distinct implementation approaches and embody different operational habits. We further conducted user study to further validate the proposed algorithm.

Ten participants (one female) aged between 22 and 27 years (average age 22.7 years) took part in the experiment. The task was two-dimensional target pursuing and obstacle avoidance as described above. The experiment involved three different control methods, which are direct human control, switching control (SWI), and SHR.

a) *Direct human control (HUM)*: The interaction force F_h was the only input to impedance model (13), and robot input u_r was zero.
b) *SWI*: Control authority alternated between human and robot. When the human exerted an interaction force, the robot input u_r was zero; otherwise, the robot followed the initially planned trajectory. At the implementation level,

we modified our proposed shared control algorithm to implement an SWI law, where replanning was avoided, and the value of PSI was 1 when interaction force was applied and 0 otherwise.

c) *SHR*: It is the algorithm proposed in this article. Considering the limited proficiency of the participants, we increased the threshold δ_f to 3.7N to reduce the likelihood of erroneous triggering of replanning. Other parameters maintained their values from Table I.

Before the experiment, we played an instructional video for the operators to help them understand the task and the three different control methods. To familiarize themselves with each control law, the subjects completed two preliminary trials for each condition. Subsequently, three sets of data were collected for each control law. After completing the experiment, each subject was asked to use a six-point Likert scale survey to evaluate their preferences for the different control methods under two subtasks: target pursuing (subtask TP) and obstacle avoidance (subtask OA). This resulted in explicit measures labeled *preference under subtask TP* and *preference under subtask OA*. The questions were as follows.

1) For the subtask of target pursuit, I prefer the HUM (or SWI, SHR) control method over the others.
2) For the subtask of obstacle avoidance, I prefer the HUM (or SWI, SHR) control method over the others.

Regarding implicit measures we evaluated:

1) the tracking errors of each target points and the average tracking error.
2) the sum of interaction force at each sample time, defined as $\sum_{t=0}^{t} ||F_t(t)||$.
3) the completion time of the task.

Then, we employed a 3 $\times$ 10 repeated measures oneway analysis of variance (ANOVA) with a significance threshold $\alpha = 0.05$ to test the null hypothesis that there is no difference between the population means for three experiment conditions. If the sphericity criterion was not met, Greenhouse–Geisser correction was applied. Upon rejection of the null hypothesis, we further conducted a post-hoc test using t-test with Bonferroni correction. The results are shown in Fig. 13 and Table II.

As shown in Fig. 13(b), the average error for HUM (M = 0.0071 m) was significantly larger than that for SWI (M = 0.0035 m) and SHR (M = 0.0032 m), while there was no significant difference between SWI and SHR. Interestingly, although the robot was entirely controlled by the operator during target pursuit under both HUM and SWI conditions, the average error for SWI was still smaller than that for HUM. One possible reason is that the desired trajectory of the robot provided a reference velocity for the operator, encouraging a slower speed of robot and resulting in higher accuracy in pursuing.

As shown in Fig. 13(c), the interaction force for SHR (M = 426.9 N) was significantly smaller than that for SWI (M = 732.7 N) and HUM (M = 947.1 N). This demonstrates that the SHR can significantly reduce human effort during task execution, which is also consistent with our design intention.

As shown in Fig. 13(d), the completion times were SHR (M = 107.89 s), SWI (M = 100.22 s), and HUM (M = 54.62 s). For HUM, the completion time is entirely determined by the

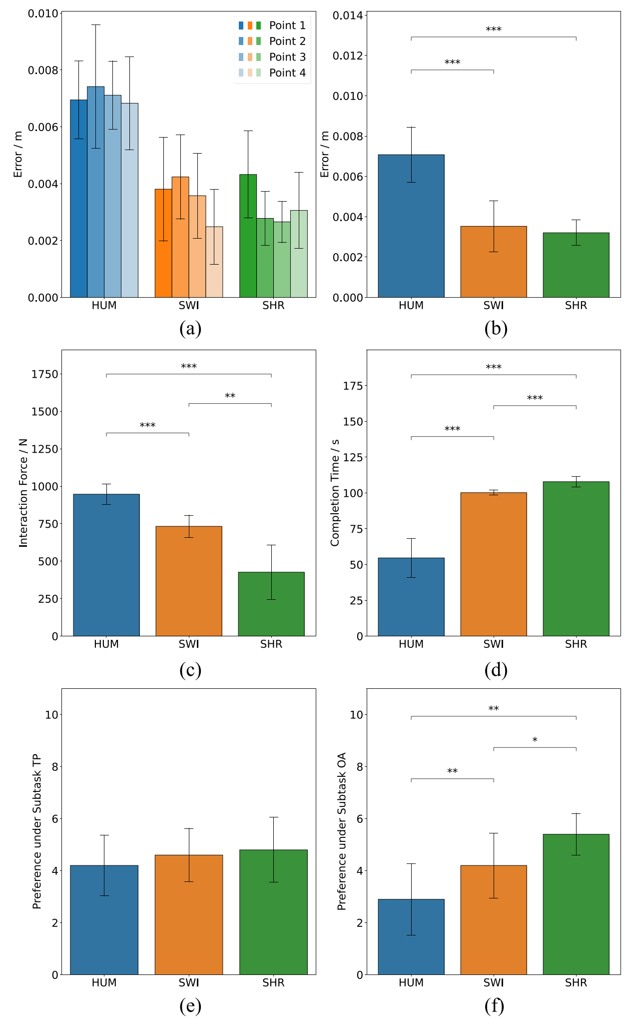

Fig. 13. Results of the user study. Each figure shows the mean and standard error of the different evaluated measures for the three control methods, where $*$ indicates $p < 0.05$, $**$ indicates $p < 0.01$, and $***$ indicates $p < 0.001$. (a) Tracking errors for each target point. (b) Average tracking error. (c) Sum of interaction forces. (d) Task completion time. (e) Preference under subtask TP. (f) Preference under subtask OA.

TABLE II
RESULTS OF POST HOC TEST ON EXPLICIT AND IMPLICIT MEASURES

Statistical analysis (one-way ANOVA, $\alpha = 0.05$)	
Average error	$(F(2, 18) = 45.300,\ p < 0.001)$
(a) > (b)	$p < 0.001$
(a) > (c)	$p < 0.001$
(b) > (c)	n.s.
Interaction force	$(F(1.220, 10.980) = 40.658,\ p < 0.001)$
(a) > (b)	$p < 0.001$
(a) > (c)	$p < 0.001$
(b) > (c)	n.s.
Completion time	$(F(1.138, 10.238) = 127.149,\ p < 0.001)$
(a) < (b)	$p < 0.001$
(a) < (c)	$p < 0.001$
(b) < (c)	$p < 0.001$
Preference under Subtask 1	$(F(2, 18) = 0.523,\ p > 0.6)$
(a) > (b)	n.s.
(a) > (c)	n.s.
(b) > (c)	n.s.
Preference under Subtask 2	$(F(2, 18) = 21.000,\ p < 0.001)$
(a) < (b)	$p < 0.01$
(a) < (c)	$p < 0.01$
(b) < (c)	$p < 0.05$

intentions and proficiency of the operators. In our experimental setup, the desired endpoints of the human and robot coincided, so participants allowed the robot to control the trajectory near the endpoint in SWI and SHR modes. In SWI mode, if the control of the robot after the handover matches the original trajectory before the endpoint, the completion time remains at the initially set 100 s, which our results confirmed. In SHR mode, although the movement time of the robot is 100 s, the additional replanning time extends the total completion time. Please note that we assumed the initial trajectory to be optimal, balancing safety and efficiency, with a preferred completion time of 100 s. Thus, deviations from this time should not be too large. While HUM can reduce completion time, it is less desirable for applications, especially those with high safety requirements.

Regarding explicit measures are shown in Fig. 13(e) and (f). No significant differences were found in preference under subtask TP among the different control methods, although SHR (M = 4.8) was numerically higher than SWI (M = 4.6) and HUM (M = 4.2). Analyzing the original data, we observed that some participants preferred the SWI method. One possible reason is that those who quickly familiarized themselves with the control method could adjust the robot's speed when pursuing targets under SWI. However, preference under subtask OA differed significantly between the control methods. The preference for SHR (M = 5.4) was higher than for SWI (M = 4.2) and HUM (M = 2.9), indicating that the SHR method effectively assists the operator in avoiding obstacles, allowing them to focus on target pursuit.

In summary, the results demonstrate that in scenarios involving target pursuit and obstacle avoidance, the proposed SHR algorithm better coordinates human–machine interactions and effectively reduces the control effort required by the human. It ensures task completion quality and enhances operational safety performance.

VI. Conclusion

In this article, we have developed a two-stage SHR framework that simultaneously adjusts the desired and actual trajectories of the robot in pHRI task, effectively addressing human–robot conflicts. When the virtual interaction force exceeds the set threshold, a real-time replanning strategy for the robot's localized desired trajectory has been designed. In cases where the interaction force is below the threshold, we introduce a shared controller that combines game theory concepts to dynamically model the human–robot relationship. In addition, we devise an index to predict system safety, which is integrated into a convex combination of the costs for both agents, ensuring that the conflict is fully resolved while maintaining safety. We evaluate the performance of the SHR algorithm through an experimental study. The proposed method is also compared with primary–secondary control strategy in a user study, which shows that our SHR algorithm yields a better performance.

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